H1. Do interior dS peaks mark genome duplication, hybridisation or assembly artefacts?

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Contents

H1.1 Interior peaks track assembly duplication 1
H1.2 The detector finds known hybrids and not old WGDs 2
H1.3 Tier A peaks are whole-genome signals 2
H1.4 Sharp peaks without BUSCO duplication (tier C) are partial annotations or partial duplications 3
H1.5 Peaks at low dS are heterozygosity (haplotype) signals, and peaks at higher dS are hybrid divergence 4
Figures 4
Reproduce 4

This report tests hypotheses about genomes whose node-dS histogram has a peak at dS <= 1.0. Method: `scripts/shape_peaks.py`. Statistics: `scripts/README.md`.

Status: interim. All numbers come from the 7,384-genome interim tables (yn00 dS). They will change when the final tables are published.

H1.1 Interior peaks track assembly duplication

Hypothesis. Genomes with an interior peak carry two diverged genome copies in the assembly, so BUSCO duplicated % is higher in them.

It would count against the hypothesis if: the share of genomes with a peak does not rise with BUSCO duplication.

Test. Share of the 5,978 evaluated genomes with an interior peak by BUSCO duplicated-% bin; Spearman correlation between BUSCO duplicated % and having a peak. Verdict rule: SUPPORTED if the top bin is more than 3 times the bottom bin and rho > 0.2.

Result.	b	genomes
<1	4349	0.0963
1-5	962	0.263
5-10	227	0.238
10-20	162	0.364
20-40	115	0.539
>=40	163	0.779

Spearman rho = 0.28 (p = 3.2e-111).

Verdict: SUPPORTED. This shows association, not mechanism. High BUSCO duplication has several causes (hybrid, dikaryon, haplotig duplication, contamination). Peaks also occur in 10% of genomes with BUSCO duplication under 1%, so BUSCO duplication is not necessary.

H1.2 The detector finds known hybrids and not old WGDs

Hypothesis. Named and literature hybrids (positive controls) usually show an interior peak. Old WGD clades (Saccharomycetaceae, Mucorales) do not, because their duplicates are older than the dS window.

It would count against the hypothesis if: most hybrid controls lack a peak, or old-WGD controls show peaks as often as hybrids.

Test. Detection rate of the shape screen in each control group. Verdict rule: SUPPORTED if hybrids are detected in >= 50% and the old-WGD rate is below half of that.

Result.	control_group	n	evaluated
Mucorales WGD	32	32	5
literature hybrid species (strain may be collapsed)	12	10	5
named hybrid (' x ' in name)	13	13	8
post-WGD Saccharomycetaceae	53	35	9
pre-WGD Saccharomycetaceae	64	12	11

Hybrid controls with a peak: 0.57. Old-WGD controls with a peak: 0.32.

Named examples:

genome	n_peaks	peak1_dS	peak1_mass	peak2_dS	busco_dup	genes_ratio_clade
Aspergillus_latus_MO46169	0				77.5	1.76
Cryptococcus_neoformans_x_Cryptococcus_deneoformans_CBS_132	1	0.331	0.666		19	1.1
Ogataea_polymorpha_x_Ogataea_parapolymorpha_CBS_1977	1	0.167	0.551		78.6	2.92
Saccharomyces_pastorianus_CBS_1484	0				51.8	2
Verticillium_longisporum_PD589	1	0.814	0.714		77.2	1.54
Zygosaccharomyces_bailii_ISA1307	1	0.137	0.779		64.1	1.97
Zygosaccharomyces_parabailii_ATCC5316	1	0.197	0.397		62.8	2.91

Verdict: INCONCLUSIVE. The old-WGD rate is not a clean false-positive rate. Only genomes with >= 300 nodes with dS > 0 are evaluated: 79 of 149 old-WGD controls pass, and the pre-WGD group keeps only 12 of 64. Pre-WGD genera such as Zygosaccharomyces and Torulaspora include hybrid strains, which are the genomes with enough paralog nodes to pass. The control set therefore mixes old-WGD and hybrid genomes. A control set that excludes known hybrid genera is needed for a real specificity estimate. Misses: *Aspergillus latus* and *S. pastorianus* have no peak at dS <= 1.0 because their peaks are above 1.0. *Z. parabailii* shows only its lower peak. The positive controls were also used to choose the thresholds, so the detection rate is optimistic.

H1.3 Tier A peaks are whole-genome signals

Hypothesis. Genomes with a sharp peak and BUSCO duplication >= 40% (tier A) hold two genome copies, so they should also have about twice the clade-median gene count and a higher share of 2-gene families.

It would count against the hypothesis if: tier A gene counts are near the clade median.

Test. Median gene-count ratio and 2-gene-family share by tier, against genomes with no interior peak. Verdict rule: SUPPORTED if tier A median gene ratio >= 1.5 times that of no-peak genomes.

Result.	tier	genomes	med_gene_ratio	frac_ratio_ge_1_5	med_2gene_share
A sharp + BUSCO dup>=40%	83	1.87	0.88	0.828	64.8
B sharp + BUSCO dup 10-40%	24	1.59	0.542	0.737	25.7
C sharp + BUSCO dup<10%	28	1.04	0.0714	0.646	4.55
D broad hump	839	1.03	0.149	0.502	1.1

Genomes with no interior peak: median gene ratio 1.02, median 2-gene share 0.50. Tier A against no-peak, gene ratio: Mann-Whitney p = 3.1e-50.

Verdict: SUPPORTED. Gene counts above the clade median also arise from annotation differences. Synteny or read depth is needed for a genome-scale claim.

H1.4 Sharp peaks without BUSCO duplication (tier C) are partial annotations or partial duplications

Hypothesis. Tier C genomes have more low gene counts than genomes without a peak, which restricts the paralog set and sharpens a small peak.

It would count against the hypothesis if: tier C low-gene-count share is similar to that of genomes without a peak.

Test. Share of tier C genomes (n = 28) with gene count below 0.7 times the clade median, against genomes with no interior peak. Verdict rule: PARTLY SUPPORTED if tier C share >= 25% and >= 2 times the no-peak share; NOT SUPPORTED otherwise.

Result. 0.32 of tier C genomes have a gene ratio below 0.7, against 0.02 of genomes with no interior peak. Median gene ratio is 1.04 in tier C and 1.02 in genomes without a peak. The majority of tier C genomes are therefore not low-gene-count.

genome	cls	peak1_dS	peak1_mass	busco_dup	genes_ratio_clade	share_2gene_fam
aff._Porphyrellus_sp._UT-M0000367_UT-M0000367	Agaricomycetes	0.405	0.728	0.4	0.165	0.467
Pulveroboletus_gigas_UT-M0000386	Agaricomycetes	0.579	0.686	0.5	0.529	0.482
Lanmaoa_rubisarca_MG1068405	Agaricomycetes	0.405	0.684	0.9	0.417	0.71
Hemileccinum_impolitum_MG1068249	Agaricomycetes	0.824	0.678	0.6	0.461	0.635
Ophiocordyceps_sinensis_CO711	Sordariomycetes	0.781	0.641	1.9	0.999	0.546
Boletaceae_sp._KM206737_KM206737	Agaricomycetes	0.201	0.639	0	0.0926	0.3
Lachancea_cidri	Saccharomycetes	0.117	0.63	4.6	1.16	0.74
Retiboletus_fuscus_TBTC15716371	Agaricomycetes	0.162	0.603	4.5	0.166	0.554
Endogone_sp._FLAS-F59071_FLAS-F59071	Endogonomycetes	0.26	0.599	0.9	1.16	0.538
Boletaceae_sp._KM206713_KM206713	Agaricomycetes	0.201	0.568	1.1	0.168	0.581
Fistulinella_prunicolor_MG1068646	Agaricomycetes	0.646	0.552	0.4	1.02	0.442
Coemansia_sp._RSA_2424_RSA_2424	Kickxellomycetes	0.297	0.544	9.4	1.45	0.742

Verdict: PARTLY SUPPORTED. A low gene ratio shows a small gene set but not the cause. For most tier C genomes the explanation is not found here; they need FCS-GX, coverage and synteny.

H1.5 Peaks at low dS are heterozygosity (haplotype) signals, and peaks at higher dS are hybrid divergence

Hypothesis. Tier A peaks are spread between heterozygosity-level dS (<= 0.1) and hybrid-level dS, and low-dS peaks are common in dikaryotic or heterokaryotic groups.

It would count against the hypothesis if: all tier A peaks sit at one dS.

Test. Distribution of tier A peak positions by phylum.

Result.	phylum	n	med_peak_dS
Ascomycota	62	0.228	0.29
Basidiomycota	16	0.153	0
Blastocladiomycota	1	0.578	0
Entomophthoromycota	1	0.199	0
Kickxellomycota	2	0.0777	1
Mucoromycota	1	0.0819	1

0.25 of tier A peaks are at dS <= 0.1 (n = 83).

Verdict: INCONCLUSIVE. dS position alone does not tell heterozygosity from hybrid divergence. Both occur in tier A. A distinction needs the parent genomes' ortholog dS or phased assemblies. Positions near 0.5 to 1.0 are lower bounds because yn00 reads low there.

Figures

Reproduce

`scripts/shape_peaks.py`, `scripts/shape_plots.py`, `scripts/make_hypothesis_reports.py`. Thresholds were set by inspecting positive controls and have not been validated on held-out data.

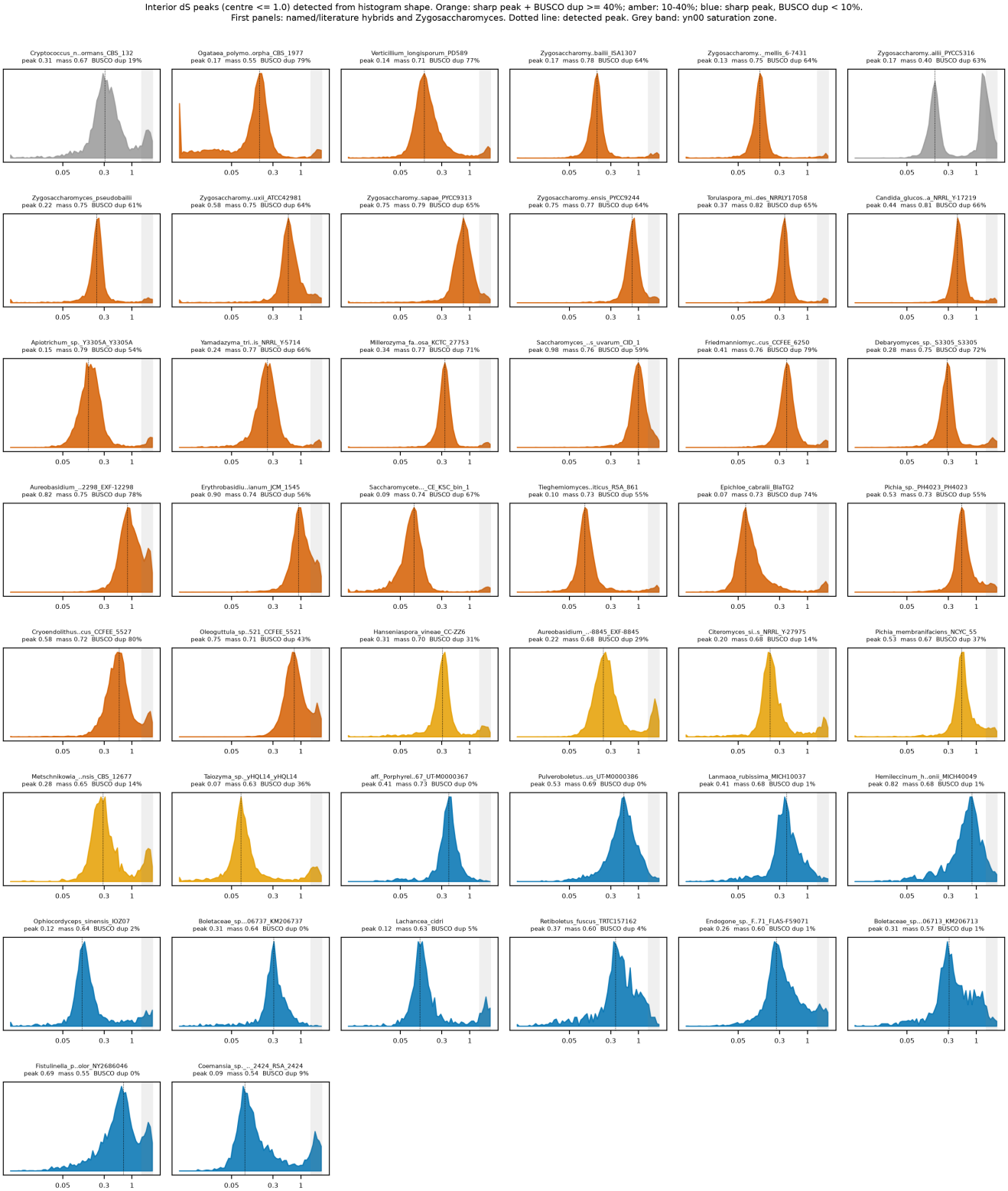


Figure 1: Shape-detected peaks

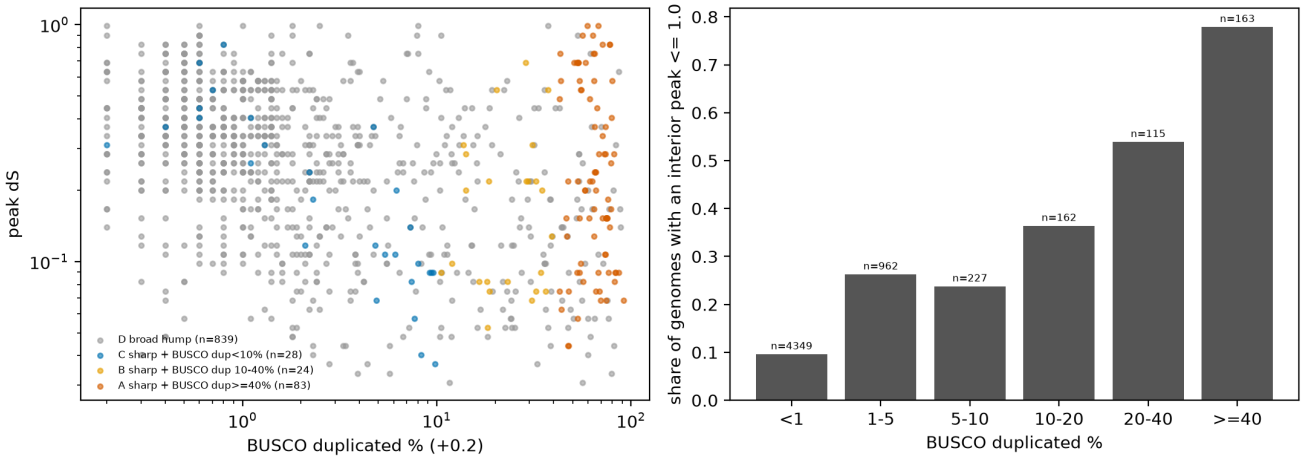


Figure 2: Peak position and BUSCO duplication